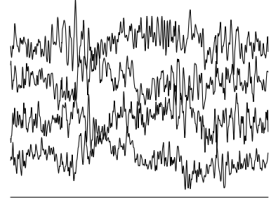
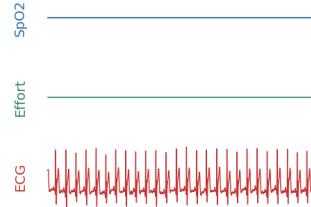


Two-stream multimodal multi-task network
 $f_{\theta} : (f^{\text{eeg}}, z^{\text{fnd}}, x^{\text{car}}) \rightarrow (\hat{y}^{\text{stg}}, \hat{y}^{\text{apn}}) \cdot 2,764,774 \text{ trainable} + 5,819,936 \text{ frozen}$



7-channel neural montage



7-channel cardiorespiratory

Expert physiology

$f^{\text{eeg}} \in \mathbb{R}^{188}$
band power, spindle, Hjorth

Frozen LaBraM

$z^{\text{fnd}} \in \mathbb{R}^{200} \cdot 5.82 \text{ M frozen}$
4 EEG channels, no gradient

Raw cardio tensor

$x^{\text{car}} \in \mathbb{R}^{7 \times 750} = 5250$
per-subject z-score, no summary

C

EEG encoder ϕ_{eeg}

FeatMLP: $388 \rightarrow e \in \mathbb{R}^{128}$
66,816 p

Cardio encoder ϕ_{car}

CardioCNN $\rightarrow c \in \mathbb{R}^{64}$
3 conv stages $\cdot$ 155,232 p

C

Cross-modal fusion

2 tokens $\rightarrow$ attention $\rightarrow z \in \mathbb{R}^{128}$
99,456 p

BiLSTM (2 layers, bidirectional)

context $L = 20 \text{ epochs} \cdot h_t \in \mathbb{R}^{512}$
2,367,488 p

Staging head

$\hat{y}_t^{\text{stg}} = \text{softmax}(W_s h_t)$
+ HMM Viterbi decode
2,565 p

Respiratory head

$\hat{y}_t^{\text{apn}} = \sigma(W_2[h_t; c_t])$
direct cardio bypass
147,969 p

direct c_t

Module detail

Feature encoder (FeatMLP)

$e = \text{GELU}(\text{LN}(W_2 \text{GELU}(\text{LN}(W_1 f))))$

Linear W_1

LayerNorm

GELU

Dropout (0.3)

Linear W_2

LayerNorm

GELU

EEG branch only: $388 \rightarrow 128$
cardio uses CardioCNN (panel C)

CardioCNN

replaces the 14 engineered cardio features
kernels sized to respiratory physiology: events
last 10–30 s, so kernels are wide and the
stack downsamples hard

Conv(7 $\rightarrow$ 48, k=25, s=2) BN GELU

MaxPool 4 $\cdot$ Dropout

Conv(48 $\rightarrow$ 96, k=15) BN GELU

MaxPool 4 $\cdot$ Dropout

Conv(96 $\rightarrow$ 96, k=7) BN GELU

concat [mean, max]

Linear $\rightarrow$ LayerNorm $\rightarrow$ GELU

5250 $\rightarrow c \in \mathbb{R}^{64}$ $\cdot$ at 25 Hz a 25-sample kernel is one second

Cross-modal fusion

tokenize: $T = [W_e e; W_c c] + M_{\text{type}}$

$\text{head}_i = \text{softmax}(Q_i K_i^T / \sqrt{d_k}) V_i$

head 1

head 2

head 3

head 4

concat heads $\cdot W_o$

+

LayerNorm

Feed-Forward $W_f \in \mathbb{R}^{128 \times 256}$

fuse $\rightarrow z \in \mathbb{R}^{128}$

$\tilde{T} = \text{LN}(T + \text{concat}_i \text{head}_i) \cdot \text{abltable: concat / eeg-only}$

Joint objective: $L = \text{CE}_{\sqrt{w}}(\hat{y}^{\text{stg}}, y^{\text{stg}}) + \lambda \text{BCE}_{\text{pw}}(\hat{y}^{\text{apn}}, y^{\text{apn}}), \lambda = 1$